MMCQA · SEMEVAL TASK

MMCultureQA

Culturally aware visual question answering in English and Arabic dialects

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Project goal

Build a system that answers open-ended questions about images that need cultural knowledge in the MENA region, in English, MSA and Arabic dialects.

The culture gap

Most vision-language models are trained on Western-centric data, so they miss MENA food, clothing, landmarks and traditions.

The language gap

Arabic comes in many dialects, such as Egyptian and Levantine, and models do worse on them than on English.

Grounded answers

Answers are free text and must be grounded in the image and fit the cultural context, so models cannot guess.

17 countries across the Middle East and North Africa

Bahrain **Egypt** **Iraq** **Jordan** **Kuwait** **Lebanon** **Libya** **Morocco** **Oman** **Palestine** **Qatar** **Saudi Arabia** **Sudan** **Syria** **Tunisia** **UAE** **Yemen**

Task and data



EXAMPLE · LEVANTINE ARABIC

شو الهدف من الإبريق النحاسي التقليدي اللي مبين بالصورة؟

الإبريق النحاسي التقليدي عادةً بيستخدم لتقديم السوائل زي الشاي، القهوة، أو المي، وغالباً بهيك مناسبات ثقافية أو رسمية.

Given an image and a question, the system produces a short, open-ended answer in the language of the question.

Semantically correct **Grounded in the image** **Right for its cultural context**

~10k training examples · 17 countries · 9 categories

- Culture, Arts & Entertainment
- Food & Cooking
- Geography, Buildings & Landmarks
- History, Geography & National Identity
- Objects, Materials & Clothing
- People, Society & Education
- Religion & Spirituality
- Sports & Recreation
- Vehicles & Transportation

Literature review: how the models are built

Paper	Input → Output	Vision encoder	Fusion	LLM
PhantomTroupe (2025)	Grayscale image + Arabic prompt → Arabic caption	Qwen2.5-VL ViT	Native projection	Qwen2.5-VL-7B (4-bit)
CultureVLM (2025)	Image + English MCQ → answer + reasoning	CLIP ViT-L/14, SigLIP	Linear / MLP	LLaVA-1.5, Phi-3-V, LLaMA-3.2-V
AIN (2025)	Image, video or doc + query → Arabic/English text	Qwen2-VL ViT	Native	Qwen2-VL-7B
Peacock (2024)	Image + Arabic prompt → Arabic text	ViT-G/14 or CLIP ViT-L/14	Q-Former or MLP	AraLLaMA-7B / AceGPT-7B
Dallah (2024)	Image + text → Arabic in matching dialect	CLIP ViT-L/14	2-layer MLP	AraLLaMA-7B

Text encoder: each model's own LLM tokenizer and embeddings (Peacock adds a BERT Q-Former).

Most models share one design: a vision encoder, a connector and an LLM. We adapt an existing model.

Literature review: training, results and gaps

Paper	Training	Dialects	Key result	Limitation	Added value
PhantomTroupe	4-bit LoRA + translation augmentation	MSA, Levantine context	Ranked 3rd (BLEU-1 22.6)	Translation errors; defaults to MSA	4-bit LoRA recipe for cultural captioning
CultureVLM	Fine-tuning on CultureVerse	English only (MENA included)	LLaVA-7B 48.1% → 90.9%	No dialect evaluation	Global cultural benchmark
AIN	Full fine-tuning, 3.6M pairs	MSA	CAMEL-Bench 63.8	MSA focus	Bilingual model, 8 domains
Peacock	Projector pretraining + LoRA	MSA; Egyptian case	VQAv2 56.2% EM	Hallucination; translation errors	Q-Former vs MLP; Henna benchmark
Dallah	3 stages incl. dialect tuning	6 dialects + MSA	Ara-LLaVA-Bench 89.7%	Misses Arab icons; dialect mixing	First six-dialect Arabic model

Dialect coverage is thin: only Dallah handles several dialects.

What's missing

Among the papers we reviewed, none combines culture, the MENA region and Arabic dialects.

Paper	Culture	MENA / Arabic	Dialects
PhantomTroupe	✓	✓	Partial
CultureVLM	✓	Partial	✗
AIN	✗	✓	✗
Peacock	Partial	✓	Partial
Dallah	Partial	✓	✓
Our project	✓	✓	✓

Each paper covers one or two of the three. We combine all three.

Implementation stages

2 of 4 stages done

1

Data preparation

Splits for English, MSA and dialects

Done ✓

2

Explore approaches

Zero-shot VLM and caption + LLM

Done ✓

3

Select the best

Compare scores and pick one approach

We are here

4

Fine-tune + evaluate

LoRA / QLoRA, then error analysis

Next

Python | PyTorch | Hugging Face | PEFT (LoRA / QLoRA) | BERTScore | Git and GitHub | Linux + GPU

Experiments: approaches compared

Task baseline

Image + question → Qwen2.5-Omni-3B (4-bit) → answer

Provided by the task creators

Direct VLM (zero-shot)

Image + question → Qwen3-VL-2B-Instruct → answer

No fine-tuning

Caption, then answer

Image → Qwen2.5-VL-3B → caption. Caption + question → LLM (4-bit) → answer

LLMs: AceGPT-v1.5 7B, Qwen3.5-4B, Jais 3 3.7B

Same setup for all: dev split · English, MSA, Egyptian, Levantine · 20 runs

BERTScore (P, R, F1)

Meaning similarity to the reference

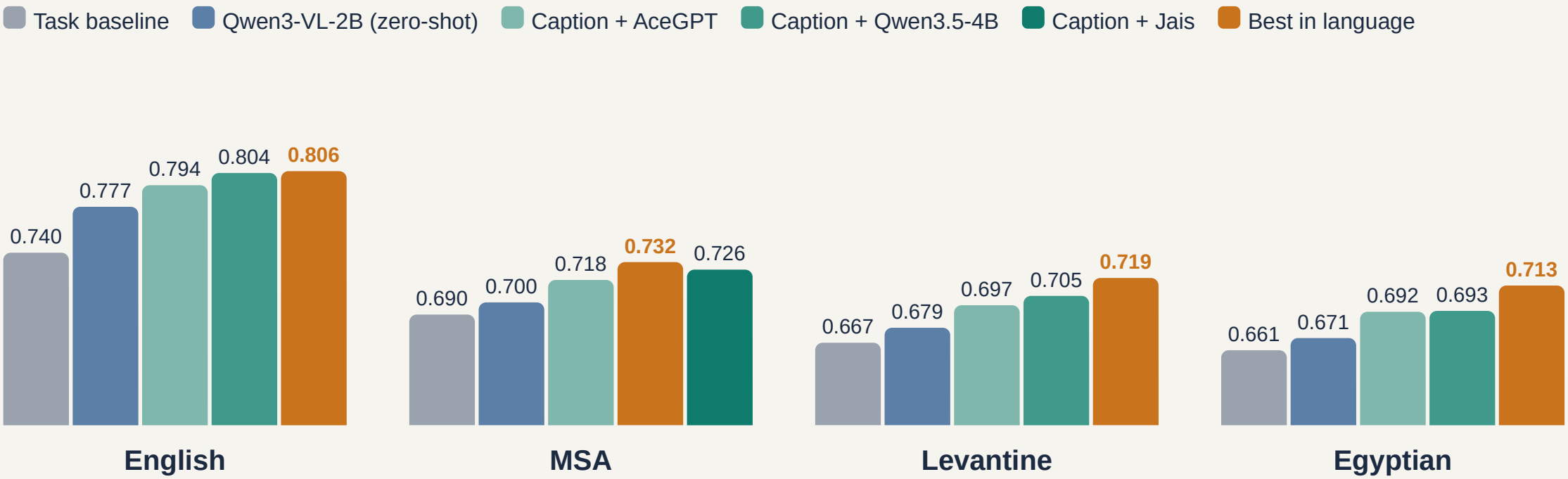
BLEU

Word overlap with the reference

ROUGE-L

Longest shared word sequence

BERTScore-F1 by approach



Caption + LLM beats the task baseline everywhere; Jais is best on dialects. Axis starts at 0.60.

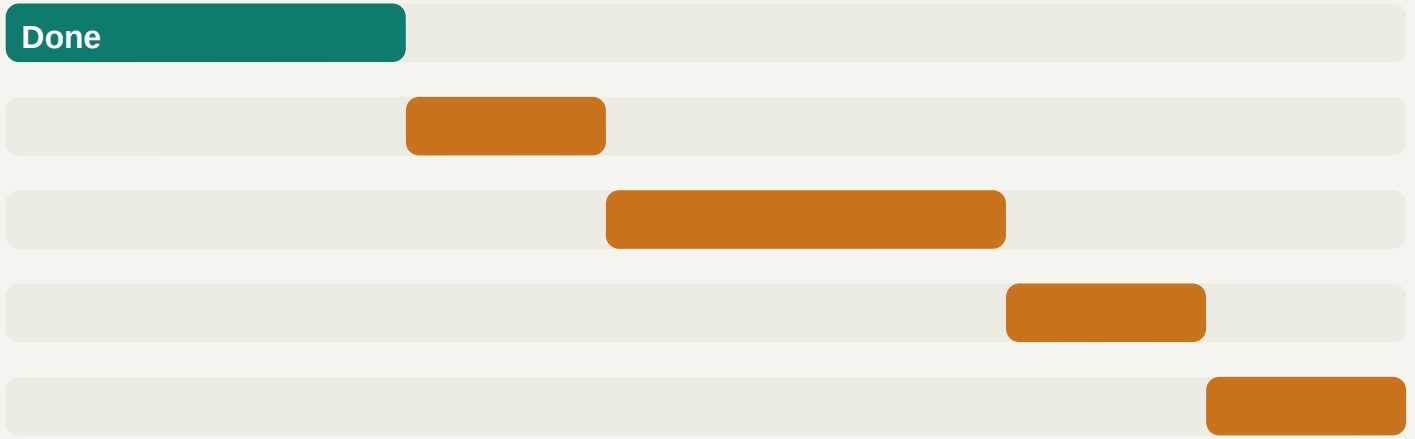
All metrics

Approach (F1 / BLEU / ROUGE-L)	English	MSA	Levantine	Egyptian
Task baseline	0.740 / 9.8 / 0.257	0.690 / 8.7 / 0.149	0.667 / 6.5 / 0.104	0.661 / 5.6 / 0.122
Qwen3-VL-2B (zero-shot)	0.777 / 17.3 / 0.360	0.700 / 8.5 / 0.161	0.679 / 7.1 / 0.110	0.671 / 6.4 / 0.141
Caption + AceGPT	0.794 / 13.6 / 0.325	0.718 / 10.4 / 0.184	0.697 / 8.0 / 0.153	0.692 / 6.6 / 0.158
Caption + Qwen3.5-4B	0.804 / 16.9 / 0.358	0.732 / 9.9 / 0.175	0.705 / 7.8 / 0.149	0.693 / 6.5 / 0.141
Caption + Jais	0.806 / 19.6 / 0.374	0.726 / 11.0 / 0.176	0.719 / 9.5 / 0.174	0.713 / 8.8 / 0.175

Dev split. Best value per metric in each language in green. BLEU and ROUGE-L are low for every approach because open-ended answers can be worded many ways.

Execution plan

1. Explore approaches
2. Choose the most promising
3. Fine-tune (LoRA / QLoRA)
4. Evaluate + error analysis
5. Report and submission



Conclusion

- Culture + MENA + dialects together is still an open gap
- Caption + LLM beats the task baseline in every language; Jais leads on dialects
- Next: choose the most promising approach and fine-tune it

Thank you · Questions?